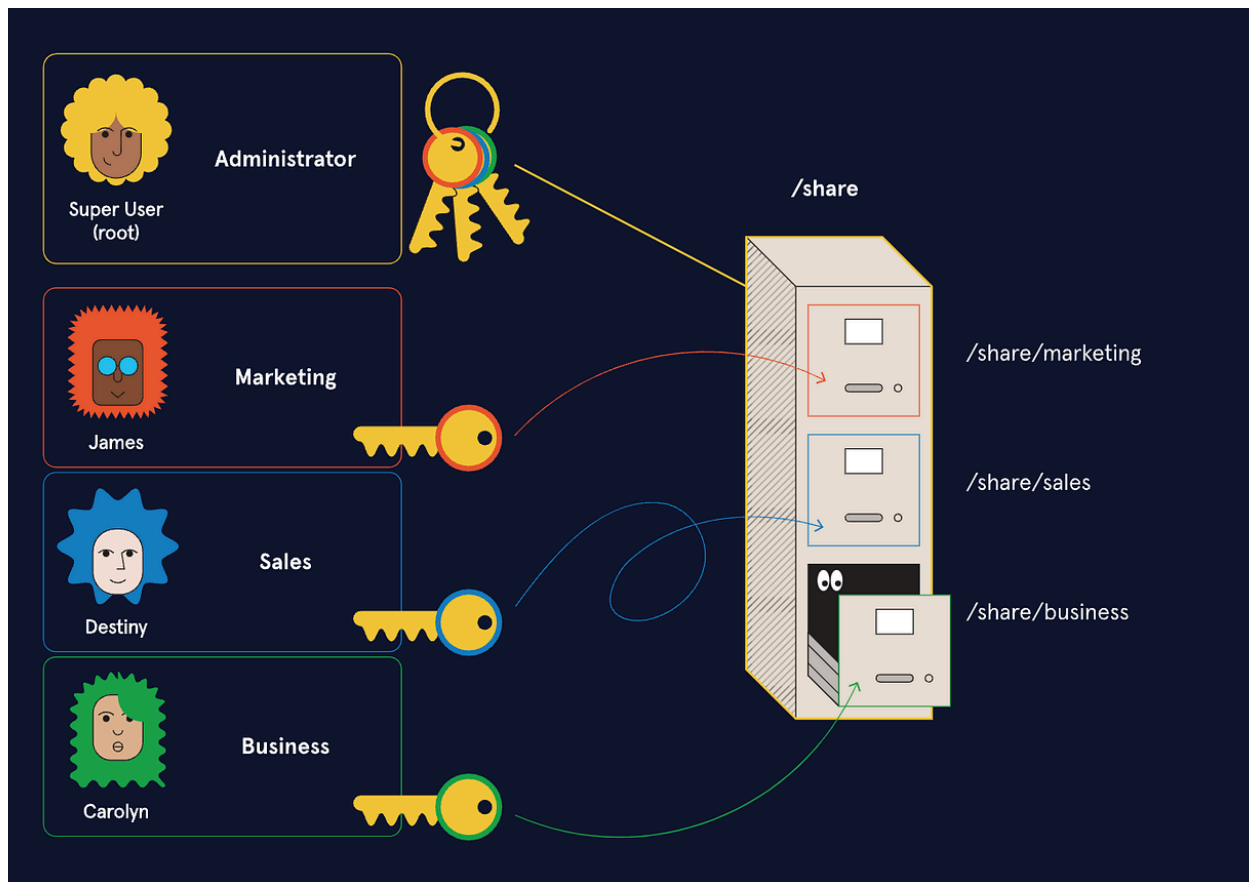


Lab 003 - Linux Users, Permissions, and Secure Administration



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Introduction

In this lab I aim to understand why one user can modify a file while another cannot. I also aim to understand how Linux controls access to files and directories.

Objectives

- Users
- Groups
- File ownership
- Permissions
- The sudo command
- Why the root account is different

Materials

1. Raspberry Pi Zero W
2. 128 GB MicroSD Card
3. Windows 11 PC/Powershell
4. Home Wi-Fi

Commands Used & Significance

- pwd (Print Working Directory) - this executes a command that tells you where you are right now.
- echo \$HOME - This evaluates a shell environment variable that tells you your designated personal folder.
 - *They are the same because, by default, Unix and Linux shells set your current working directory to your home directory the moment you open a new terminal window.
- sudo -l - this command shows what your account is allowed to do with elevated privileges.
- r - Read: Allows you to view the contents of a file or list the files inside a folder.
- w - Write: Allows you to create, edit, or delete a file or directory.
- x - Execute: Allows you to run a file (like a script or program) or "enter" and navigate through a folder (using the `cd` command).
- chmod 444 - remove write permission

- nano - allows editing the file
 - *I could not save changes because I did not have write permission.
- chmod 644 - restores write access
- cat - displays the contents of a file in the terminal.
- chmod - change mode
- + - add a permission
- touch - is to create an empty file if it doesn't already exist.

CONCLUSION

During this lab I explored how Linux controls access through users, groups, ownership, and permission bits. By intentionally removing write permissions with **chmod 444**, I observed how applications such as **nano** respect filesystem permissions. Restoring permissions with **chmod 644** reinforced how Linux security is enforced through ownership and permission settings.

Screenshots

```

student@pi-lab: ~/permissions_lab
student@pi-lab:~$ whoami
student
student@pi-lab:~$ id
uid=1000(student) gid=1000(student) groups=1000(student),4(adm),20(dialout),24(cdrom),27(sudo),29(audio),44(video),46(plugdev),60(games),100(users),102(netdev),986(gpio),988(l2c),989(spi),992(render),996(input),1001(lpadm)
student@pi-lab:~$ pwd
/home/student
student@pi-lab:~$ echo $HOME
/home/student
student@pi-lab:~$ sudo -l
usage: sudo -h | -e | -k | -V
usage: sudo -v [-Abiks] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-Abiks] [-g group] [-h host] [-p prompt] [-U user]
           [-e user] [command [arg ...]]
usage: sudo [-Abiks] [-r role] [-t type] [-C num] [-D directory]
           [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
           [-e user] [Mk=value] [-i | -s] [command [arg ...]]
usage: sudo -e [-Abiks] [-r role] [-t type] [-C num] [-D directory]
           [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
           [-e user] file ...
student@pi-lab:~$ sudo -l
Matching Defaults entries for student on pi-lab:
    env_reset, mail_badpass, secure_path=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin, use_pty, env_keep=NO_AT_BRIDGE,
    env_keep=DPKG_DEB_THREADS_MAX, timestamp_type=global, env_keep="http_proxy HTTP_PROXY", env_keep="https_proxy HTTPS_PROXY",
    env_keep="ftp_proxy FTP_PROXY", env_keep="RSYNC_PROXY", env_keep="no_proxy NO_PROXY"
User student may run the following commands on pi-lab:
    (ALL : ALL) ALL
student@pi-lab:~$ ALL
-bash: ALL: command not found
student@pi-lab:~$ ALL : ALL
-bash: ALL: command not found
student@pi-lab:~$ pwd
/home/student
student@pi-lab:~$ mkdir permissions_lab
student@pi-lab:~$ cd permissions_lab
student@pi-lab:~/permissions_lab$ touch confidential.txt
student@pi-lab:~/permissions_lab$ ls -l
total 0
-rw-r--r-- 1 student student 0 Jul 1 18:02 confidential.txt
student@pi-lab:~/permissions_lab$ chmod 444 confidential.txt
student@pi-lab:~/permissions_lab$ ls -l
total 0
-r--r--r-- 1 student student 0 Jul 1 18:02 confidential.txt
student@pi-lab:~/permissions_lab$ nano confidential.txt
student@pi-lab:~/permissions_lab$ chmod 444 confidential.txt
student@pi-lab:~/permissions_lab$ ls -l
total 0
-r--r--r-- 1 student student 0 Jul 1 18:02 confidential.txt
student@pi-lab:~/permissions_lab$ nano confidential.txt
student@pi-lab:~/permissions_lab$ chmod 644 confidential.txt
student@pi-lab:~/permissions_lab$ nano hello.sh
student@pi-lab:~/permissions_lab$ ls -l
total 4
-rw-r--r-- 1 student student 0 Jul 1 18:02 confidential.txt
-rw-r--r-- 1 student student 47 Jul 1 18:11 hello.sh
student@pi-lab:~/permissions_lab$ ./hello.sh
-bash: ./hello.sh: Permission denied
student@pi-lab:~/permissions_lab$ chmod +x hello.sh
student@pi-lab:~/permissions_lab$ ls -l

```

```
student@pi-lab:~/permissions_lab
total 0
-rw-rw-r-- 1 student student 0 Jul 1 18:02 confidential.txt
student@pi-lab:~/permissions_lab $ chmod 444 confidential.txt
student@pi-lab:~/permissions_lab $ ls -l
total 0
-rw-rw-r-- 1 student student 0 Jul 1 18:02 confidential.txt
student@pi-lab:~/permissions_lab $ nano confidential.txt
student@pi-lab:~/permissions_lab $ chmod 444 confidential.txt
student@pi-lab:~/permissions_lab $ ls -l
total 0
-rw-rw-r-- 1 student student 0 Jul 1 18:02 confidential.txt
student@pi-lab:~/permissions_lab $ nano confidential.txt
student@pi-lab:~/permissions_lab $ chmod 644 confidential.txt
student@pi-lab:~/permissions_lab $ nano confidential.txt
student@pi-lab:~/permissions_lab $ nano hello.sh
student@pi-lab:~/permissions_lab $ ls -l
total 4
-rw-rw-r-- 1 student student 0 Jul 1 18:02 confidential.txt
-rw-rw-r-- 1 student student 47 Jul 1 18:11 hello.sh
student@pi-lab:~/permissions_lab $ ./hello.sh
-bash: ./hello.sh: Permission denied
student@pi-lab:~/permissions_lab $ chmod +x hello.sh
student@pi-lab:~/permissions_lab $ ls -l
total 4
-rw-rw-r-- 1 student student 0 Jul 1 18:02 confidential.txt
-rwxrwxr-x 1 student student 47 Jul 1 18:11 hello.sh
student@pi-lab:~/permissions_lab $ ./hello.sh
Hello from my Raspberry Pi!
student@pi-lab:~/permissions_lab $ sudo ls /root
student@pi-lab:~/permissions_lab $ ls /root
ls: cannot open directory '/root': Permission denied
student@pi-lab:~/permissions_lab $ chmod+x ls /root
-bash: chmod+x: command not found
student@pi-lab:~/permissions_lab $ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:mailing list Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd/bin/false
systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin
messagebus:x:990:990:systemd-busctl:/nonexistent:/usr/sbin/nologin
sshd:x:989:65534:sshd user:/run/ssh:/usr/sbin/nologin
dnsmasq:x:988:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:101:104:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/usr/sbin/nologin
student@pi-lab:~/permissions_lab $ ls -ld /etc
```